1 SHEAR

1.1 CONCRETE CONTRIBUTION V_c

Primary (ACI 318 simplified):

$$V_c = 2\lambda\sqrt{f'_c}\, b_w\, d \tag{1}$$

λ lightweight concrete modification factor, 1.0 normal weight
f'_c specified concrete compressive strength [psi]
b_w web width [in]
d effective depth to tension steel centroid [in]

LIMITS

Stirrups required	when $V_u > 0.5\,\phi V_c$

1.2 STEEL CONTRIBUTION V_s

$$V_s = \frac{A_v\, f_{yt}\, d}{s} \tag{2}$$

A_v area of shear reinforcement within spacing s [in^2]
f_{yt} yield strength of transverse reinforcement [psi]
s stirrup spacing along member axis [in]

LIMITS

Upper cap on V_s	$V_s \le 8\sqrt{f'_c}\, b_w d$
Min stirrup area	$A_{v,\min} = \max\left(0.75\sqrt{f'_c}\,\dfrac{b_w s}{f_{yt}},\ \dfrac{50\, b_w s}{f_{yt}}\right)$
Max spacing, $V_s \le 4\sqrt{f'_c}\, b_w d$	$s_{\max} = \min(d/2,\ 24 \text{ in})$
Max spacing, $V_s > 4\sqrt{f'_c}\, b_w d$	$s_{\max} = \min(d/4,\ 12 \text{ in})$

1.3 COMBINED

$$\phi V_n = \phi(V_c + V_s) \ge V_u, \qquad \phi = 0.75 \tag{3}$$

2 MOMENT

2.1 STRESS BLOCK DEPTH a

$$a = \frac{f_y A_s}{0.85 f'_c b} \tag{4}$$

f_y yield strength of longitudinal tension reinforcement [psi]
A_s area of longitudinal tension reinforcement [in^2]
f'_c specified concrete compressive strength [psi]
b width of compression face [in]

2.2 NOMINAL MOMENT CAPACITY

$$\phi M_n = \phi f_y A_s \left(d - \frac{a}{2}\right), \qquad \phi = 0.9 \tag{5}$$

2.3 REQUIRED STEEL AREA $A_{s,\text{req}}$

Combining a and $\phi M_n \ge M_u$ and solving the quadratic in a:

$$A_{s,\text{req}} = \frac{0.85 f'_c b}{f_y}\left[d - \sqrt{d^2 - \frac{2M_u}{0.85\,\phi\, f'_c\, b}}\right] \tag{6}$$

M_u factored moment demand [lb-in]

3 DEFLECTION

3.1 CRACKING MOMENT M_{cr}

$$M_{cr} = \frac{f_r\, I_g}{y_t}, \qquad f_r = 7.5\lambda\sqrt{f'_c} \tag{7}$$

f_r modulus of rupture [psi]
I_g gross (uncracked) moment of inertia [in^4]
y_t distance from centroid to extreme tension fiber [in]

3.2 CRACKED TRANSFORMED SECTION I_{cr}

Modular ratio and concrete modulus:

$$n = \frac{E_s}{E_c}, \qquad E_c = 57{,}000\sqrt{f'_c}\ \text{[psi]} \tag{8}$$

Neutral axis depth kd from first moments about the NA (tension concrete ignored):

$$\frac{b(kd)^2}{2} = nA_s(d - kd) \tag{9}$$

Cracked transformed moment of inertia:

$$I_{cr} = \frac{b(kd)^3}{3} + nA_s(d - kd)^2 \tag{10}$$

n modular ratio, steel to concrete
E_s steel modulus of elasticity, typically 29,000 ksi
E_c concrete modulus of elasticity [psi]
kd depth from extreme compression fiber to neutral axis [in]

3.3 EFFECTIVE MOMENT OF INERTIA I_e (BRANSON)

$$I_e = \left(\frac{M_{cr}}{M_a}\right)^3 I_g + \left[1 - \left(\frac{M_{cr}}{M_a}\right)^3\right] I_{cr} \tag{11}$$

M_a maximum service moment at the stage deflection is computed [lb-in]